

Khulna University, Khulna
 Computer Science and Engineering Discipline
 4th Year, 1st Term Examination 2024
 Session: 2022-2023
 Course No: CSE-4103
 Full Title of Course: **Computer Graphics**

Date: 12/05/2024

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Full Marks: 60

Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section

Section A

There are FOUR questions in this section. Answer any THREE questions

- Q1. a) Explain, how do digital image processing differ from computer graphics? 2.5
- b) What is an image aspect ratio? Can a $5 \times 3\frac{1}{2}$ inch image be presented at 6×4 inch without introducing geometric distortion? Explain why or why not. 2.5
- c) If you want to cut a 512×512 sub-image out from the center of an 800×600 image, what are the coordinates of the pixel in the large image that is at the lower left corner of the small image? 1.5
- d) Find the CMY coordinates of a color at (0.2, 1, 0.5) in the RGB space. 1.5
- e) Is overstrike is harmful besides wasting time? 02

- Q2. a) What is incremental scan-conversion method? 01
- b) In the Bresenham's circle algorithm the decision parameters are: 02

$$d_{i+1} = \begin{cases} d_i + 4x_i + 6 & \text{if } d_i < 0 \\ d_i + 4(x_i - y_i) + 10 & \text{if } d_i \geq 0 \end{cases}$$

However, in the mid point circle algorithm the decision parameters are:

$$d_{i+1} = \begin{cases} d_i + 2x_i + 3 & \text{if } d_i < 0 \\ d_i + 2(x_i - y_i) + 5 & \text{if } d_i \geq 0 \end{cases}$$

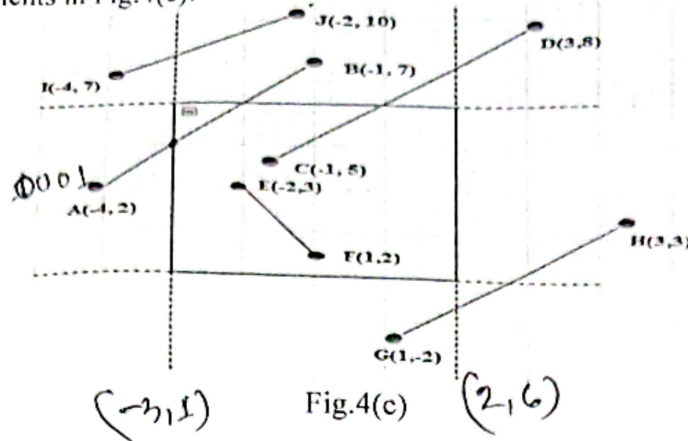
Do the both methods give the same circle or not? Explain.

- c) Why are the mid point ellipse algorithm calculate two decision parameters for the first quadrant? How can we eliminate overstrike? 03
- d) What are the steps required to plot a line whose slope is between 90° and 135° using Bresenham's algorithm? 04

- Q3. a) Perform a 45° rotation of triangle A(0, 0), B(1, 1), C(5, 2) about the point B. Then modify the transformed triangle to twice its size. 04
- b) Describe the transformation M_L which reflects an object about a line L. Also show that reflection about the line $y=x$ is attained by reversing coordinates. That is $M_L(x, y)=(y, x)$. 04
- c) Explain phong specular light reflection model. 02

- Q4. a) Find the coordination under which $S_{S_x, S_y} \cdot R_\theta = R_\theta \cdot S_{S_x, S_y}$ 03
- b) Find the workstation transformation that maps the normalized devices screen onto a physical device whose x extent is 0 to 199 and y extent is 0 to 639. Where the origin is located at the (i) lower left corner and (ii) upper left corner of the device. 03
- c) Left R be the rectangular window whose lower left-hand corner is L(-3,1) and upper right-hand corner is at R(2,6). Find the region code for the end points and the clipping categories 04

for the line segments in Fig.4(c).



Section B

There are FOUR questions in this section. Answer any THREE questions

- Q5. a) Determine whether a point $P(x, y)$ lies to the left or to the right of a line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$? 02
- b) Suppose that in an implementation of the Cohen-Sutherland algorithm we choose boundary lines in the top-bottom-right-left order to clip a line in category 3, draw a picture to show a worst-case scenario, i.e., one that involves the highest number of iterations. 02
- c) Use the Weiler-Atherton algorithm to clip the polygon in Fig. 5(c) 02

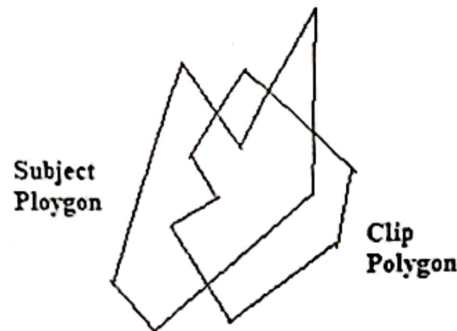
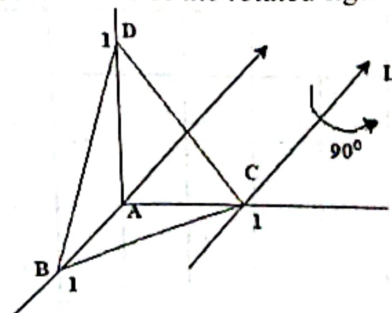


Fig. 5(c)

- d) A rotation is performed about the x axis followed by a rotation about the y axis or a rotation is performed about the y axis followed by a rotation about the x axis. Does the order of performing the rotation matter? 03
- e) How many sub divisions are required to clip a line segment with M pixels using the mid point subdivision process. 01
- Q6. ~~7~~ The pyramid defined by the coordinates $A(0, 0, 0)$, $B(1, 0, 0)$, $C(0, 1, 0)$ and $D(0, 0, 1)$ is rotated 90° about the line L that has the direction $V=J+K$ and passing through point $C(0, 1, 0)$ given in Fig 6(a). Find the coordinates of the rotated figure. 04



- b) Find the matrix for mirror reflection with respect to the plane passing through the point (1, 1, 1) and having a normal vector whose direction is $N=I+J+K$. 04
- c) Show that the alignment transformation satisfies the rotation $A_V^{-1} = A_V^T$. 02

- Q7. a) Define projection. What are the types of projection? What are applications of each type? 03
- b) Under the standard transformation Per k, where the view plane is the xy plane and the center of projection is C (0, 0, -d), what is the projected image of a point in the plane $z=-d$. 03
- c) Derives the perspective transformation onto a plane with reference point $R_0(2,2,2)$, normal vector $N=2I+2J+2K$, and using C(1,1,1) as the centre of projection. 04

- Q8. a) Why are hidden-surface algorithms needed? Why is it easier to locate hidden surfaces when parallel projection is used? 02
- b) What is the underlying concept of the painter's or the priority algorithm? 01
- c) Describe how hidden surface removed and projection are integrated into the ray-fracing process. 02
- d) Describe the one-principle-vanishing perspective and two-principle-vanishing point perspective. 03
- e) State how Gouraud surface rendering method works. 02

Date: 16/05/2024

Khulna University, Khulna
Computer Science and Engineering Discipline
4th Year, I Term Examination
Session: 2022-2023
Course No: BA 4151
Full Title of Course: Accounting

Full Marks: 60

Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

Section A

There are **FOUR** questions in this section. Answer any **THREE** questions

- ✓ Q1. a) Briefly describe the three core activities in the accounting process. 03
- b) "Remaining responsible to the third parties means accountability." Based on the statement describe "Accounting in accountability". 05
- ✓ c) What is the *historical cost principle*? 02
- ✓ Q2. Morning Star Company Ltd. has the following transactions during March, 2024: 10
- 01.03.2024: The owner invested TK. 500,000 in the business
- 03.03.2024: Purchased supplies on credit TK. 5,000
- 08.03.2024: Provide services and received TK. 12,500 from customer
- 11.03.2024: Insurance expense paid TK. 2000
- 15.03.2024: Interest paid TK 1000
- 20.03.2024: Purchased an equipment costing TK. 30,000, of which TK. 10,000 was paid in cash
- 28.03.2024: Paid salary of employees TK. 20,000

✓ Prepare Journal for the above transactions.

- ✓ Q3. a) List down the limitations of a Trial Balance. 03
- b) Prepare a Trial Balance from the following ledger balances of **Tesco Enterprise**: 07

✓ Accounts Receivable	7,640
Accounts Payable	8,390
✓ Cash	12,480
✓ Equipment	49,360
✓ Gasoline Expense	750
✓ Utilities Expense	520
Notes Payable	17,000
✓ Prepaid Insurance	1,960
✓ Maintenance and Repair Expenses	960
Service Revenue	10,610
✓ Owner's Drawings	700
Owner's Capital	1,22,000
✓ Salaries and Wages Expense	4,420
Salaries and Wages Payable	810
✓ Rent Expense	12,000

Unearned Service Revenue	20,000
✓Supplies	25,000
✓Furniture	90,000
Interest Payable	780
Accumulated Depreciation	1,200
Bank Loan	25,000

- Q4. a) RTL Inc. began operations on April 1, 2024. At the end of April 2024, management wants to prepare monthly financial statements for the month of April 2024. The following information relates to April. 08
- On April 01, 2024 the firm took a loan of Tk. 6,00,000 at 12% interest per year. The interest is payable after a year.
 - On April 01, the firm paid Tk. 60,000 for a 12 month insurance policy and debited as prepaid insurance.
 - Two employees of the firm get a monthly salary of Tk. 10,000 each. Salary of every month is paid on the 10th day of next month.
 - On April 01, Tk. 100,000 was received from client for 10 months car rent in advance and recorded as Unearned Rent Revenue.
- Prepare the adjusting entries needed for month ended on April 30, 2024.
- b) Explain any two Features of a Transaction. 02

Section B

There are **FOUR** questions in this section. Answer any **THREE** questions

- Q5. Following is the Adjusted Trial Balance of Tech Sound Company on December 31 2023. 10

Tech Sound Company Ltd. Trial Balance December 31 2023

Account Title	Debit (TK)	Credit (TK)
Cash	150,000	
Prepaid Insurance	12,000	
Owners' Capital		135,000
Supplies	10,000	
Building	120,000	
Accumulated Depreciation		13,500
Rent Revenue		12,000
Accounts Receivable	22,500	
Accounts Payable		30,600
Interest expense	5,500	
Utility Expense	3,500	
Service Revenue		42,800
Supplies expense	6,500	
Salaries Expense	7,500	
Salaries and wage payable		5,500
Depreciation Expense	2,500	
Drawing	7,000	
Insurance Expense	3,000	
Unearned Service Revenue		8,800
10% Notes Payable		100,000

Interest payable		1,800
Total	350,000	350,000

Prepare an Income Statement and Balance Sheet from the above information.

- Q6. a) What do you mean by Prime Cost in a cost sheet? 02
 b) The following cost and inventory data for the last year are taken from the accounting records of D4Tech who manufactures and sells Computer keyboard: 08

Costs incurred:

Beginning Inventory of raw material	2,25,000	Advertisement	80,000
Purchase of raw material	13,75,000	Maintenance, Factory	25,000
Ending Inventory of raw material	4,00,000	Sales Commission	1,50,000
Direct Labor	1,25,000	Salesperson Salary	1,75,000
Utilities, Factory	7,80,000	Indirect labor	1,30,000
Depreciation, Factory	3,60,000	Office Rent	1,35,000
Insurance, Factory	30,000	Office Depreciation	40,000
Supplies, Factory	50,000	Office Supplies	20,000

The company wants to make a 40% profit margin.

Total Unit of Production and sales is 20,000 units.

Prepare a cost sheet and show Total Manufacturing Cost, Total Selling Price and Per Unit Selling price.

- Q7. a) What do you understand by Variable cost and Fixed cost? Explain with examples. 04
 b) A company makes Keyboards with a selling price of Tk. 435 per unit and variable costs of Tk. 335 per unit. The fixed cost for the period is Tk. 10,00,000. 02
 i) How many Keyboards do they need to sell to reach break-even point? 02
 ii) What is Break-even point Sales Volume in Taka? 02
 iii) What is the required output level to make a target profit of Tk. 4,00,000? 02

- Q8. Networks Company manufactures wireless routers. The company's contribution format income statement for the most recent year is given below:

	Total	Per Unit	Percentage of Sales
Sales (2500 units).....	\$2,50,000	\$100	100%
Less variable expenses.....	1,50,000	60	? %
Contribution margin.....	1,00,000	\$ 40	? %
Less fixed expenses.....	80,000		
Operating income.....	20,000		

- i. Compute the company's break-even point in both units and sales dollars. Use the equation method. 02
 ii. Assume that sales increase by \$60,000 next year. If cost behaviour patterns remain unchanged, by how much will the company's operating income increase? 02
 iii. The marketing manager argues that a \$90,000 increase in the monthly advertising budget would increase monthly sales and total sales will be \$450,000. Should the advertising budget be increased? 03
 iv. Refer to the original data. Management is considering using higher-quality components that would increase the variable cost by \$10 per unit. The marketing manager believes the higher-quality product would increase sales by 50% per month. Should the higher-quality components be used? 03

Khulna University
Computer Science and Engineering Discipline
4th Year Term I Examination 2024

Session: 2022-2023

Course No: CSE 4105

Title of Course: Computer Security

Full Marks: 60

Time: 03 Hours

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- Use separate sheet for each section.

Section A

There are **FOUR** questions in this section. Answer any **THREE** questions.

1. (a) What are the three basic components of computer security? 3
 (b) What is threat? Define the type of attack in each of the following class: 5
 - (i) A student breaks into a professor's office to obtain a copy of the next class test.
 - (ii) A student gives a check for 100 to buy a used book. Later the student finds out that the check was cashed for 1000.
 - (iii) A student sends hundreds of emails per day to the university using a phony return email address.
- (c) Define plain text, cipher text, encryption and decryption. 2
2. (a) How many keys are required for two people to communicate via a cipher? 2
 (b) Is the set of integer a group with respect to multiplication? Justify your answer. 3
 (c) Prove that $(a-b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$. 3
 (d) Find the $\gcd(2420, 230)$ using prime factorization. 2
3. The problem provides a numerical example of encryption using a single round version of DES (without the initial permutation, inverse initial permutation and permuted choices). We start with the bit pattern $\{FFFFCCCAA9999\}_H$ for the plaintext and $\{FFF00000FFF\}_H$ for the round subkey. [Use Table 1-3 given at the end.] 10
 - (i) Derive L_0 and R_0 .
 - (ii) Find the value $F(R_0)$
 - (iii) Write down the ciphertext after first round.
4. (a) In a public-key system using RSA, you intercept the ciphertext $C = 146$ sent to the user whose public key is $e = 7$ and $n = 187$. What is the value of plaintext M ? 6
 (b) What is Euler's Totient function? In case of Euler's Totient function, prove that $\phi(n) = \phi(p) \times \phi(q)$, where $n = pq$, p and q both are prime numbers. 4

Section B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) Discuss the distribution process of public keys using public key certificates with appropriate figure. 4
 (b) Users Alice and Bob use the Diffie-Hellman key exchange technique with a common prime $q = 29$ and a primitive root $\alpha = 10$. 6
 - (i) If Alice has private key $X_{\text{Alice}} = 13$, what is Alice's public key Y_{Alice} ?

- (ii) If Bob has private key $X_{\text{Bob}} = 17$, what is Bob's public key Y_{Bob} ?
 (iii) What is the shared secret key?

6. (a) What is digital signature? What are the requirements of a digital signature scheme? 4
 (b) Consider $G = Z_{29}^*$ with order $d=28$ and $g=3$. If the signer's first private key $x=7$ and second private key $k=11$. Show the signing and verification for message $m=27$ using ElGamal Digital signature scheme. 6

7. (a) What is HMAC (hash based Message Authentication Code)? Describe the structure of HMAC with a proper diagram. 6
 (b) List four attack strategy to discover a password and their countermeasures to keep secure the password. 4

8. (a) Discuss DAC and RBAC process with appropriate figures. 6
 (b) What is firewall? What are the functions performed by IDS? 4

Table 1: Expansion Permutation (E)

32	01	02	03	04	05
04	05	06	07	08	09
08	09	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	01

Table 2: Permutation Function (P)

16	7	20	21	29	12	28	17
1	15	23	26	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

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Table 3: Definition of DES S-Boxes

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
S_1	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S_2	15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
	3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
	0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
	13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S_3	10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
	13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
	13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
	1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S_4	7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
	13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
	10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
	3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S_5	2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
	14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
	4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
	11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S_6	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

S_7	4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
	13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
	1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
	6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S_8	13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
	1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
	7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
	2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

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Khulna University, Khulna
Computer Science and Engineering Discipline
4th Year, 1st Term Examination
Session: 2022-2023
Course No: SOC-4153
Full Title of Course: Government and Sociology

Full Marks: 60

Time: 03 Hours

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

Section A

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | |
|-----|--|------|
| Q1. | a) What do you understand by Politics? | 02 ✓ |
| | b) Critically analyze the functions of modern state with example? | 08 ✓ |
| Q2. | a) What is local government? | 02 ✓ |
| | b) Demonstrate the present situation of government and politics in Bangladesh. | 08 ✓ |
| Q3. | a) Explain the salient features of socialism. | 04 |
| | b) Describe the characteristics of major administrative systems of developed countries. | 06 |
| Q4. | How does local government contribute to grassroots development and democracy in Bangladesh? Provide examples to support your answer. | 10 ✓ |

Section B

There are **FOUR** questions in this section. Answer any **THREE** questions

- | | | |
|-----|---|------|
| Q5. | a) Define Sociology. | 03 ✓ |
| | b) As a student of Computer Science and Engineering Discipline, why would you study sociology? Explain. | 07 ✓ |
| Q6. | a) What is urban ecology? | 03 |
| | b) Discuss the problems and prospects of urbanization in the context of Bangladesh. | 07 |
| Q7. | a) What do you understand by culture and civilization? | 04 ✓ |
| | b) Briefly explain the relationship between Sociology and Political Science. | 06 ✓ |
| Q8. | a) What is industrial revolution? | 02 ✓ |
| | b) State the basic characteristics of the social structure of Bangladesh. | 08 ✓ |

Khulna University, Khulna
Computer Science and Engineering Discipline
4th Year, 1st Term Examination
Session: 2022-2023
Course No: CSE-4121

Full Title of Course: **Applied Probability and Queuing Theory**

Full Marks: **60**

Time: **03 Hours**

- The figures in the margin indicate full marks. The questions are of equal value.
- Use separate sheet for each section.

Section A

There are **FOUR** questions in this section. Answer any **THREE** questions

- Q1. a) What do you mean by probability? Show that $P(E|F) = \frac{P(EF)}{P(F)}$. 03
- b) A prisoner has got a peculiar death sentence. According to the sentence, he will roll a dice 10 times at most. If he gets six for at least 5 times, he will be freed. The prisoner starts playing. Determine the following possibilities: 04
- (i) What is the probability that he will stop playing after his eighth round realizing that he will not be freed?
- (ii) What is the possibility that he will be freed after his eighth round?
- c) A laboratory blood test is 98% effective in detecting a certain disease when it is, in fact, present. However, the test also yields a "false positive" result for 1% of the healthy person tested. If 0.5% of the population actually has the disease, what is the probability a person has the disease given that his test result is positive? 03
- Q2. a) An experiment consists of tossing a fair coin and then rolling a *four-sided die (i.e., the face values of the die are 1, 2, 3, and 4)*. The face value of the coin is considered 0 (tail) or 1 (head). Two random variables, X and Y, have been defined in the following way. 05
- $X = \text{The face value of the die} + 2$
- $Y = (\text{Face value of the die} - \text{Face value of the coin})^2$
- By determining the co-variance between X and Y show that they are positively correlated.
- b) In an exam, two students (A and B) are answering the same set of five multiple choice questions. Each question has 4 alternatives of which only one is the correct answer. A student will have to select one option from each question. Consider that every student guesses at the answer. What is the possibility that A will have one more correct answer than B? 05
- Q3. a) Explain conditional probability with an example. 02
- b) Bill and George go target shooting together. Both shoot at a target at the same time. Suppose Bill hits the target with probability 0.7, whereas George, independently, hits the target with probability 0.4. 04
- (i) Given that exactly one shot hit the target, what is the probability that it was George's shot?
- (ii) Given that the target is hit, what is the probability that George hit it?
- c) In answering a question on a multiple-choice test a student either knows the answer or guesses. Let p be the probability that she knows the answer and 1 - p the probability that she guesses. Assume that a student who guesses at the answer will be correct with probability 1/m, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that she answered it correctly? 04

- Q4. a) What do you mean by probability mass function and probability density function?
b) Suppose that $p(x, y)$, the joint probability mass function of X and Y , is given by

$$p(1, 1) = 0.2, p(1, 2) = 0.1, p(1, 3) = 0.2, p(2, 1) = 0.1, p(2, 2) = 0.1, \\ p(3, 2) = 0.2, p(3, 1) = 0.1$$

Calculate the conditional probability mass function of X given that $Y > 1$.

- c) Suppose, RT PCR machines for COVID 19 test is 90 percent effective in detecting Corona disease when it is really present. However, the test also shows a "false positive" result for 3 percent of the healthy persons tested (if a healthy person is tested, then, with probability 0.03, the test result will imply s/he has Corona disease). If 15 percent of the population actually has Corona disease, what is the probability a person has the Corona disease given that his/her test result is positive?

Section B

There are **FOUR** questions in this section. Answer any **THREE** questions

- Q5. a) Let us assume that a particular day's weather can be sunny (s), cloudy (c), or rainy (r). Tomorrow's weather condition depends only on today's weather condition in the following way. 05
- (i) If it is sunny (s) today, then tomorrow will be sunny (s), cloudy (c), or rainy (r) with probabilities: 0.6, 0.3, 0.1 respectively.
 - (ii) If it is c today, then tomorrow will be s, c, or r with probabilities: 0.3, 0.4, 0.3 respectively.
 - (iii) If it is r today, then tomorrow will be s, c, or r with probabilities: 0.1, 0.2, 0.7 respectively.
- Suppose that it is raining today. What is the probability that the day after tomorrow will be cloudy?
- b) An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.6 is the same color, and with probability 0.4 is the opposite color, as the ball it replaces. If initially one ball is red and the other is blue, find the probability that the third ball selected is red. 05
- Q6. a) Let us assume that a three state Markov chain has the following single step transition probabilities. 06
- $$\begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.4 & 0.2 \end{bmatrix}$$
- Determine the long run proportions that the chain will spend in the three states.
- b) Specify the classes of the following Markov chain. Determine whether the *classes are transient or recurrent*. 04
- $$\begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$
- Q7. a) Suppose that customers arrive at a Poisson rate of one per every 20 minutes, and that the service time is exponential at a rate of one service per 10 minutes. Determine the values of L and W? 04
- b) What do you mean by a M/M/3 queuing model? Write down the balance equations for such a model and determine the equation for P_n in terms of λ , μ , n, and P_0 . The terms have their usual meanings. 06
- Q8. a) What do you mean by open queuing network and closed queuing network? 02
- b) Suppose, traffic to a message switching center for one of the outgoing communication lines arrive in a random pattern at an average rate of 240 messages per minute. The line has a transmission rate of 800 characters per second. The message length distribution (including control characters) is approximately exponential with an average length of 176 characters. Calculate the following assuming that a very large number of message buffers are provided: 08
- (i) Average number of messages in the system
 - (ii) Average number of messages in the queue waiting to be transmitted.
 - (iii) Average time a message spends in the system.
 - (iv) Average time a message waits for transmission

The following cost and inventory data for the last year are taken from the accounting records of D4Tech who manufactures and sells Computer keyboard:

Costs incurred:

Beginning Inventory of raw material	2,25,000	Advertisement	80,000
Purchase of raw material	13,75,000	Maintenance, Factory.	25,000
Ending Inventory of raw material	4,00,000	Sales Commission	1,50,000
Direct Labor	1,25,000	Salesperson Salary	1,75,000
Utilities, Factory	7,80,000	Indirect labor	1,30,000
Depreciation, Factory	3,60,000	Office Rent	1,35,000
Insurance, Factory	30,000	Office Depreciation	40,000
Supplies, Factory	50,000	Office Supplies	20,000

The company wants to make a 40% profit margin.

Total Unit of Production and sales is 20,000 units

Prepare a cost sheet and show:

Total Manufacturing Cost

Total Selling Price

Per Unit Selling price

1. SST Traders has the following transactions during the month of January 2024.

<u>Date</u>	<u>Transaction</u>
January 01	The owner invested Tk. 5,00,000 cash in the business.
January 05	Purchased equipment for Tk. 1,00,000, paying 50,000 in cash and the rest on account.
January 07	Purchased Inventory on credit from Apex Traders of Tk. 2,00,000.
January 10	Goods sold for Tk. 1,30,000 and received cash Tk. 1,00,000.
January 31	The firm paid Tk. 20,000 cash for store rent for January and also paid Tk. 50,000 as salary for January.

Prepare journal entries for the above transactions.

2. Prepare a Trial Balance from the following ledger balances of AKS Enterprise:

Accounts Receivable	7,642	Prepaid Insurance	1,968
Accounts Payable	8,396	Maintenance and Repair Expenses	961
Cash	12,481	Service Revenue	10,610
Equipment	49,360	Owner's Drawings	700
Gasoline Expense	758	Owner's Capital	42,000
Utilities Expense	523	Salaries and Wages Expense	4,428
Notes Payable	17,000	Salaries and Wages Payable	815